CAPSTONE PROJECT • CENTRAL EUROPEAN UNIVERSITY × ILF CONSULTING ENGINEERS

🇺🇿 Uzbekistan — Power Sector Transition Tracker

Forecasting electricity demand to 2040 · scenario analysis · investment signals · regional pipeline

📊 **Forecast winner: Prophet** (CV MAPE 6.4%) 🎯 **2030 target:** 54% renewables ⚡ **Horizon:** 2024 – 2040 🔄 **Refreshed:** May 2026

SCENARIO

- ○ BAU
- ◉ Government
- ○ Accelerated

DEMAND-SIDE EFFICIENCY (%)

0% 10% 20% 30%

T&D LOSSES (%)

4% 7% 10%

PLAN B — NUCLEAR

☐ Include nuclear

NUCLEAR CAPACITY (MW)

0 1.2 GW 2.4 GW 3.6 GW

COMMISSION YEAR

'30 '32 '34

SECTIONS

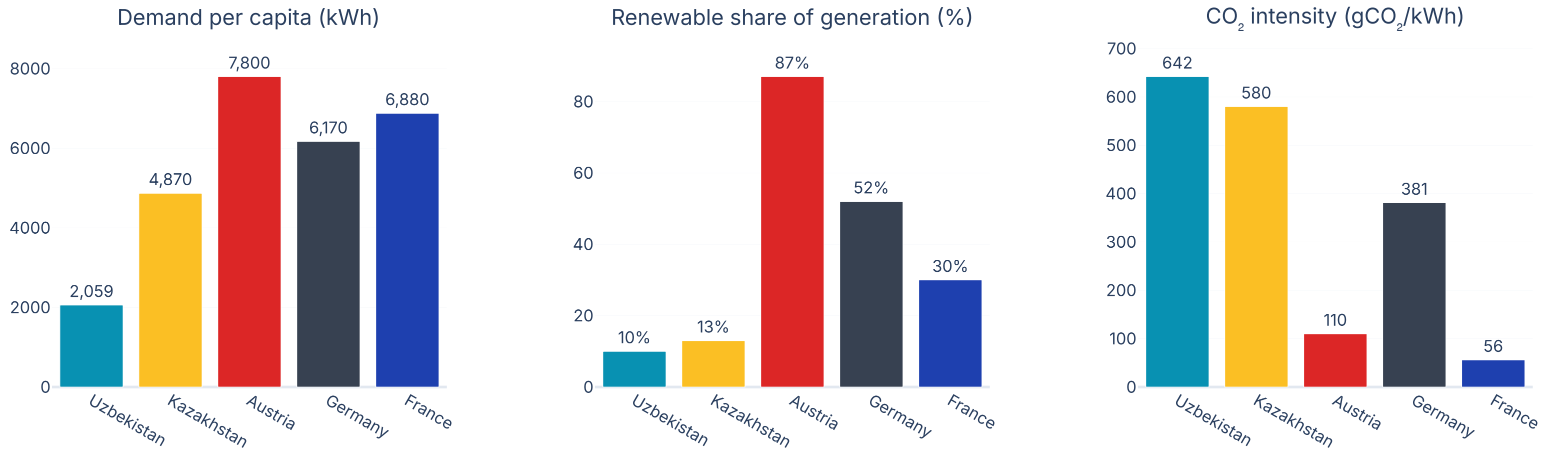
- 🌍 Global benchmarks
- 🇺🇿 Country snapshot
- 🧭 Scenario definitions
- 📈 Demand forecast
- ⚖️ Supply–Demand Balance
- ⚡ Generation mix
- CO₂ emissions

🌍 Global benchmarks — where Uzbekistan stands

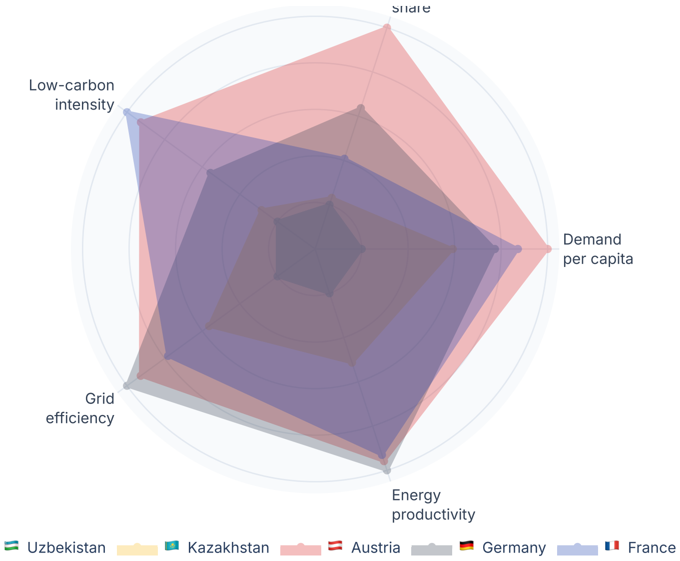
Uzbekistan vs Kazakhstan (CIS peer), Austria (renewables-led), Germany (energy transition leader), and France (nuclear-led). All 2023 figures. Scroll the chart to compare electricity demand, RE share, and CO_2 intensity side-by-side.

UZBEKISTAN	KAZAKHSTAN	AUSTRIA	GERMANY	FRANCE
2,059	**4,870**	**7,800**	**6,170**	**6,880**
kWh/capita · 10% RE · 642 gCO_2/kWh	kWh/capita · 13% RE · 580 gCO_2/kWh	kWh/capita · 87% RE · 110 gCO_2/kWh	kWh/capita · 52% RE · 381 gCO_2/kWh	kWh/capita · 30% RE · 56 gCO_2/kWh

Five-country comparison (2023)



Power-system performance — radar (higher = better on every axis)



🇺🇿 Country snapshot

Headline indicators for the latest confirmed year — and where the selected scenario takes us by 2030 and 2040.

DEMAND 2023
⚡ **73.4**
TWh → 103 (2030), 143 (2040)

RE SHARE 2030 (GOVERNMENT)
♻️ **48.6%**
2040: 52.0% · target 54%

POWER CO_2 2030
34.0 Mt
intensity 302 gCO_2/kWh

PLAN B NUCLEAR
⚛️ **0.0 GW**
off

🧭 Scenario definitions

Three policy-anchored scenarios + parametric Plan B nuclear. All anchored to end-2025 actuals (~5.6 GW solar+wind commissioned).

BAU

CURRENT BUILD PACE CONTINUES

~60% of the official 2030 RE target met. No additional policy push beyond what is already financed.

2030 capacity: ~13 GW total RE
2030 RE share: ~36% RE
2040 RE share: ~32% RE
2040 CO_2: ~73 Mt
Capex: $32 bn (2024-2040)

Risk: Misses 54% target by 18 pp. Supply-deficit risk under high-demand path.

Government ● Active

HITS THE OFFICIAL APRIL 2025 TARGETS

12 GW solar + 8 GW wind + 4.7 GW hydro by 2030. Tracks Uzbekistan's Concept of Decarbonisation 2030.

2030 capacity: ~25 GW total RE
2030 RE share: ~49% RE
2040 RE share: ~52% RE
2040 CO_2: ~33 Mt
Capex: $56 bn (2024-2040)

Risk: Short of 54% target by ~5 pp — Plan B nuclear closes it.

Accelerated

STRETCH — EXCEEDS OFFICIAL TARGETS

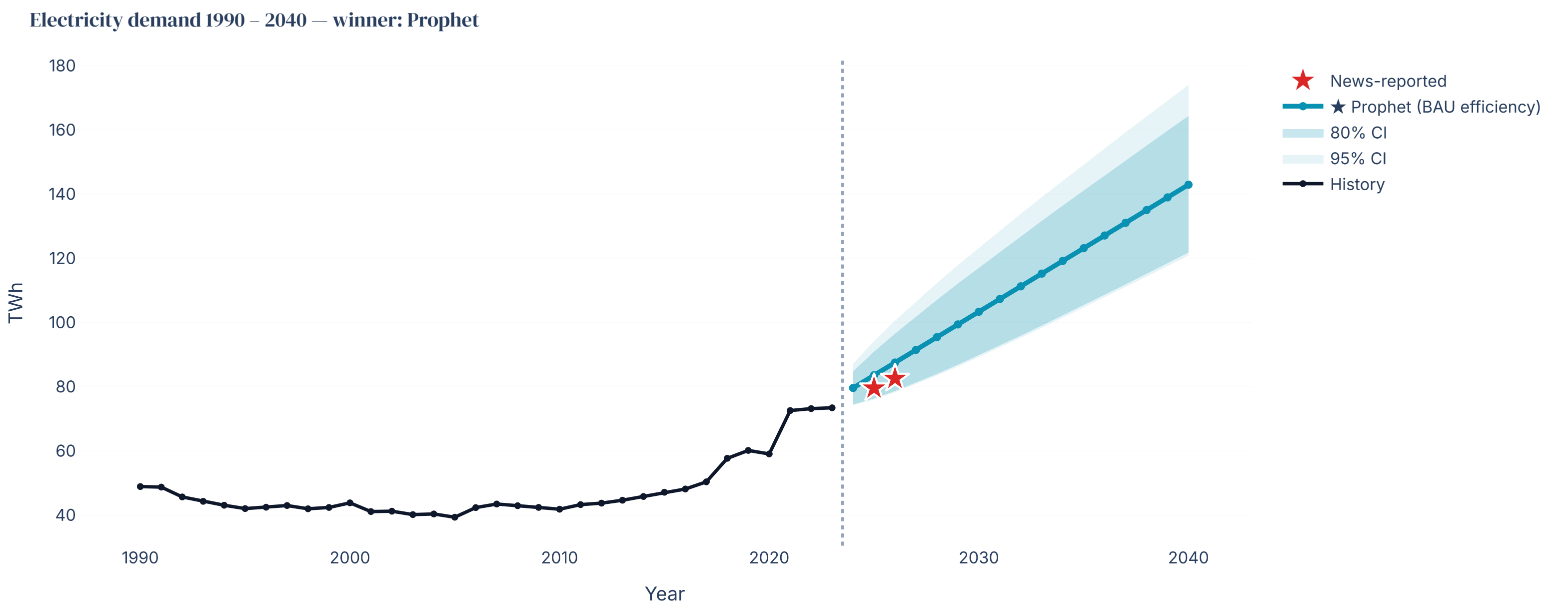
15 GW solar + 10 GW wind + 5.2 GW hydro by 2030. Front-loads investment; assumes faster permitting and transmission.

2030 capacity: ~30 GW total RE
2030 RE share: ~59% RE
2040 RE share: ~69% RE
2040 CO_2: ~21 Mt
Capex: $76 bn (2024-2040)

Risk: Most aggressive RE build any CIS country would have attempted; supply chain & grid integration are the constraints.

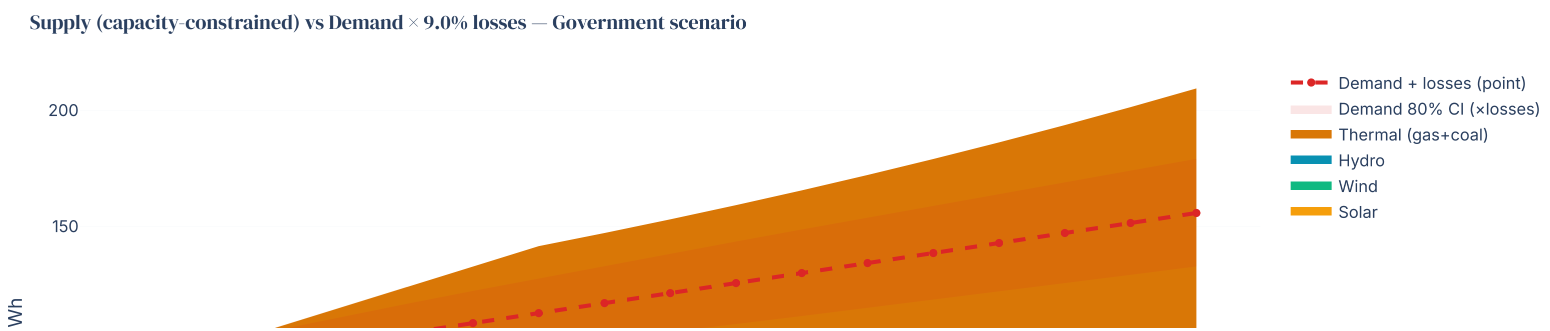
📈 Electricity demand forecast

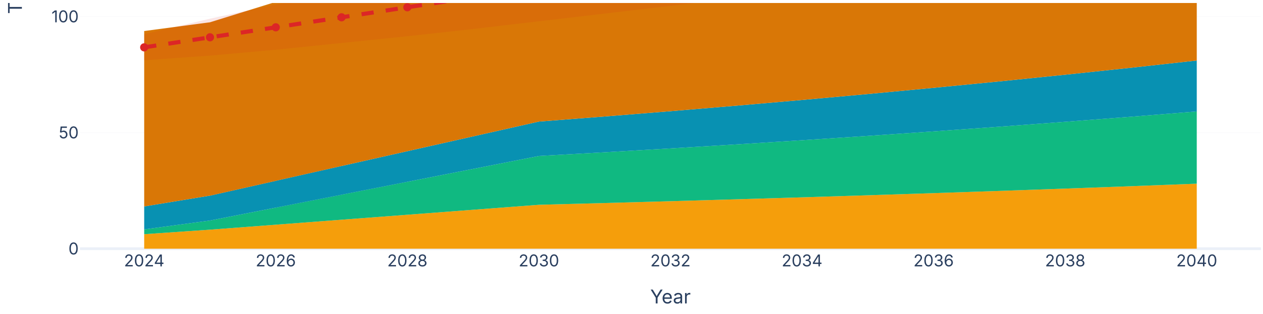
Eleven models compared; winner = Prophet (CV MAPE 6.4%). 80% / 95% bootstrap CIs shaded. Red stars are news-reported actuals. The dashed orange line shows demand after applying your demand-side efficiency slider.



⚖️ Supply–Demand Balance

Capacity-constrained supply (capacity × annual CF) vs demand × (1 + losses). Red bands flag deficit years where supply cannot cover demand. Move the Demand-side efficiency and T&D losses sliders in the sidebar to close the gap.





✅ **2030 SUPPLY SURPLUS**

Surplus = 28.9 TWh — supply 141 TWh exceeds demand+losses 113 TWh.

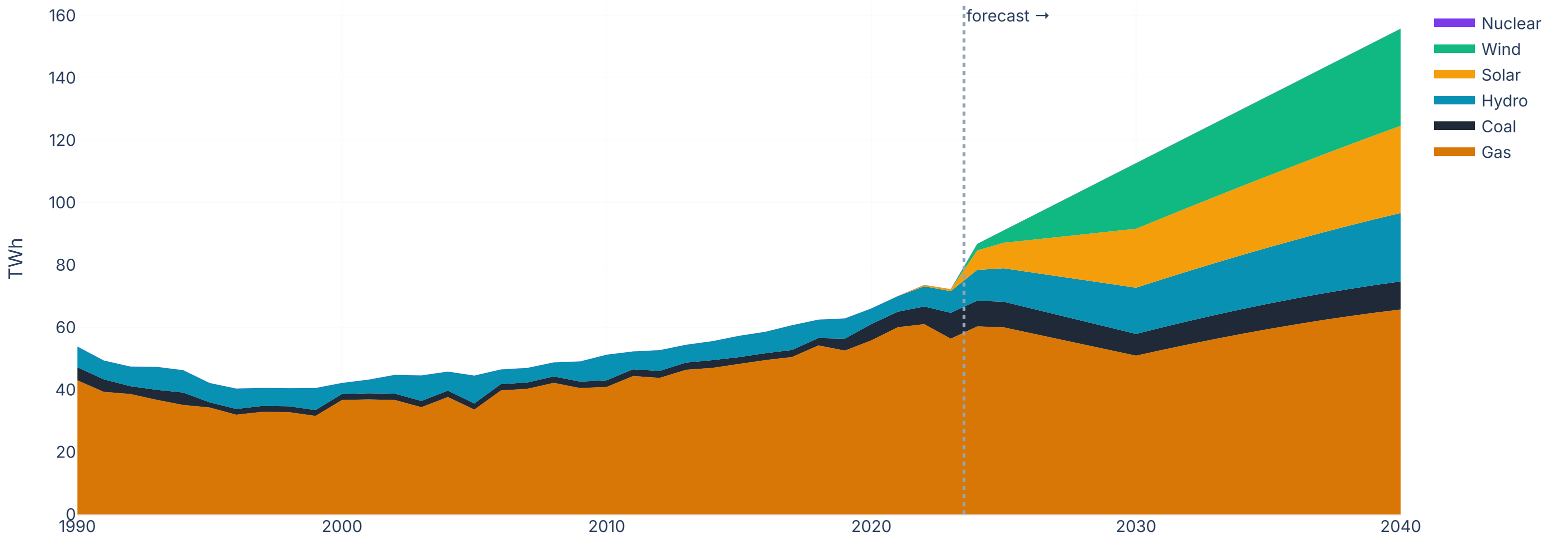
Strategic levers under surplus:

- Electricity exports to Afghanistan, Pakistan (CASA-1000 corridor) or EU via CASPER.
- Earlier coal retirements without supply risk.
- Green hydrogen production (sink the variable RE surplus).
- Curtailment savings via more storage — advisory opportunity.

⚡ Generation mix by scenario

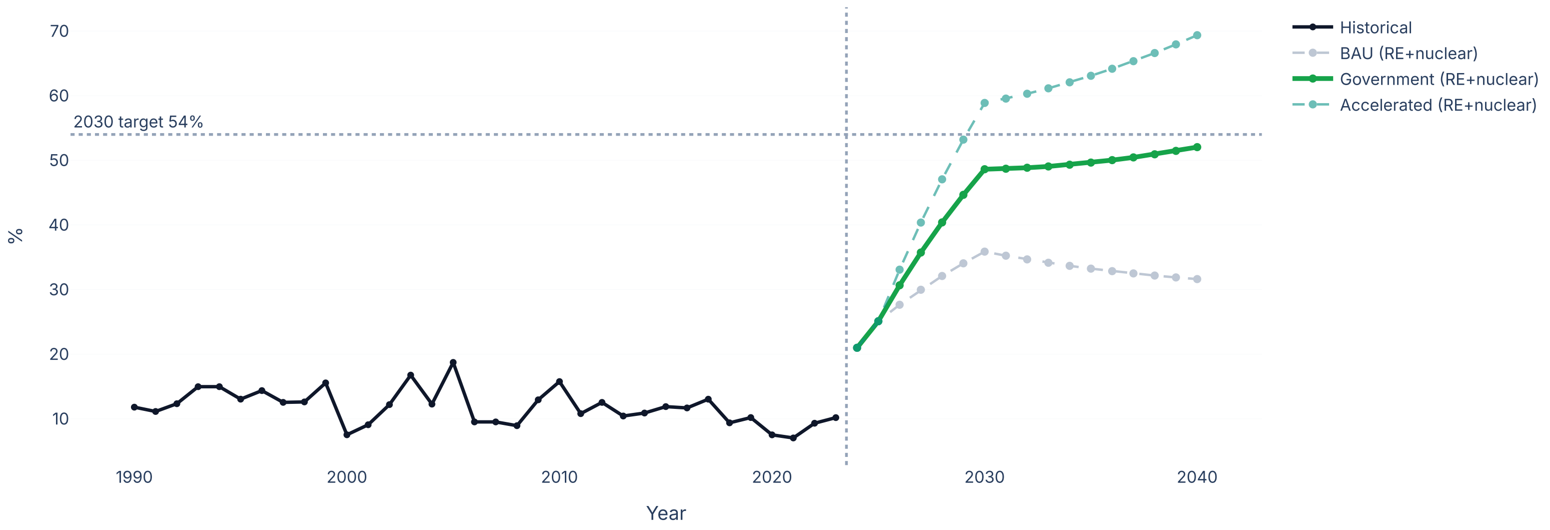
Toggle scenarios + Plan B nuclear in the sidebar; charts re-compute live.

Generation mix — Government scenario



Year

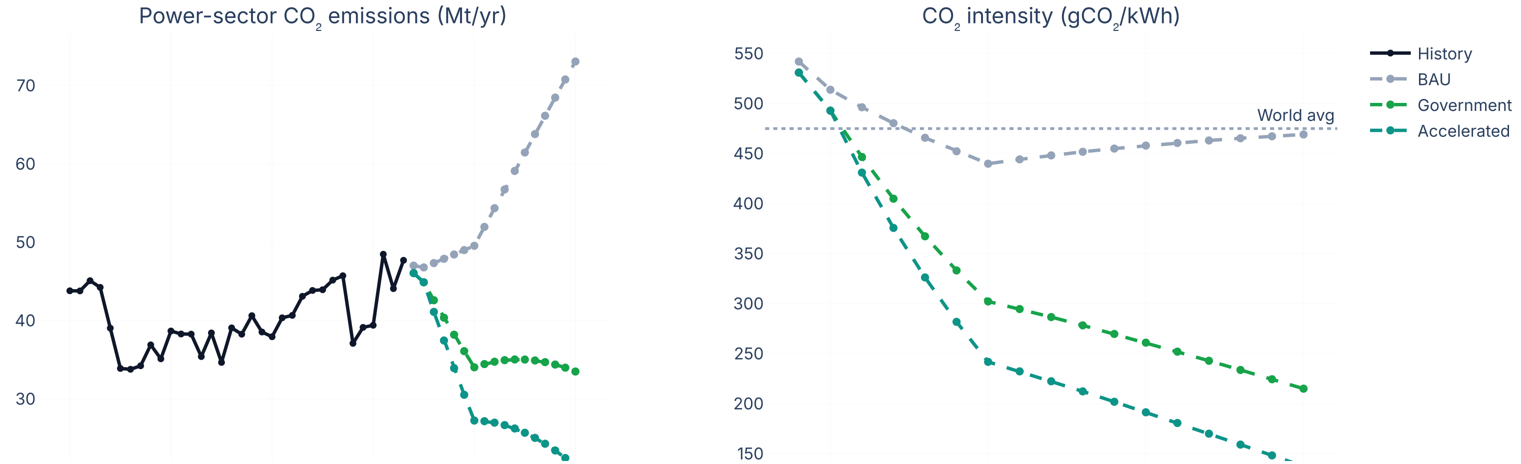
Low-carbon share of generation — all scenarios



CO_2 emissions outlook

Power-sector emissions across the three scenarios, with intensity benchmarked to world average.

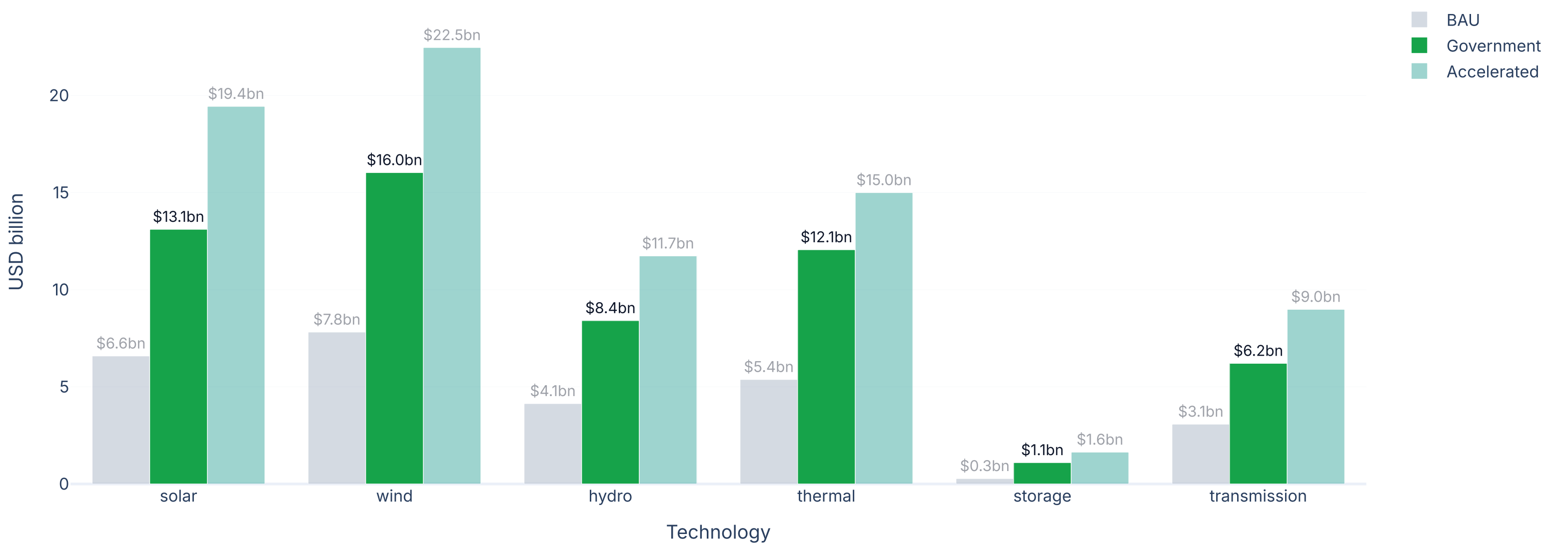
Power-sector carbon footprint



💰 Investment outlook 2024 – 2040

Implied capex by technology and scenario, using IRENA / IEA unit-cost ranges. Highlighted bars match your sidebar selection.

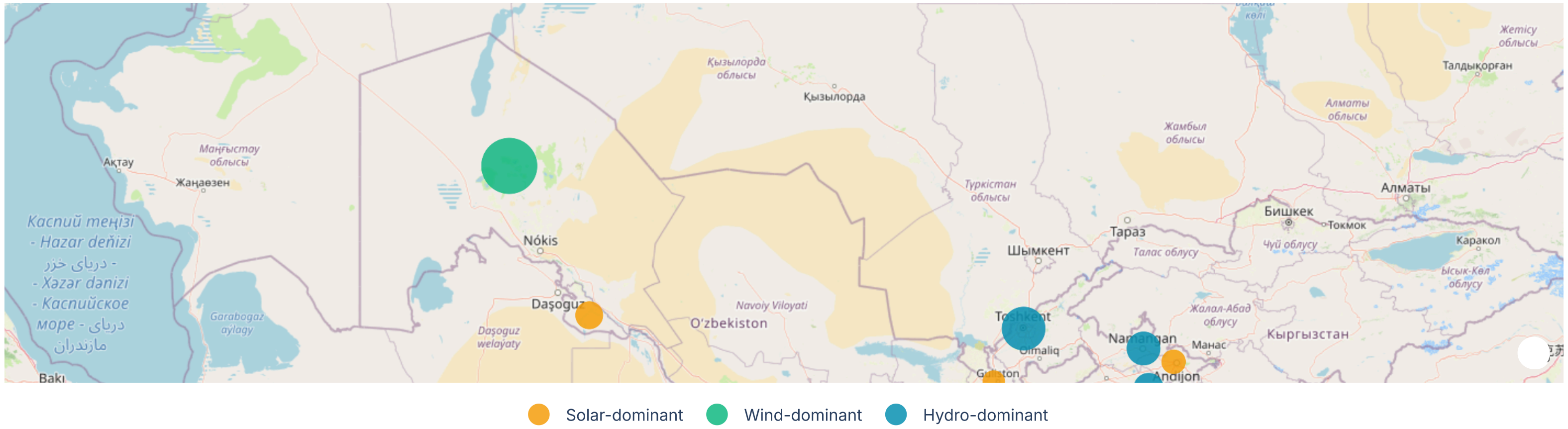
Capex by technology & scenario (2024 – 2040, USD bn)



🗺️ Regional renewable pipeline — by oblast

13 oblasts mapped. Bubble area ∝ total RE capacity. Tap a marker for project list. Tiles: OpenStreetMap.

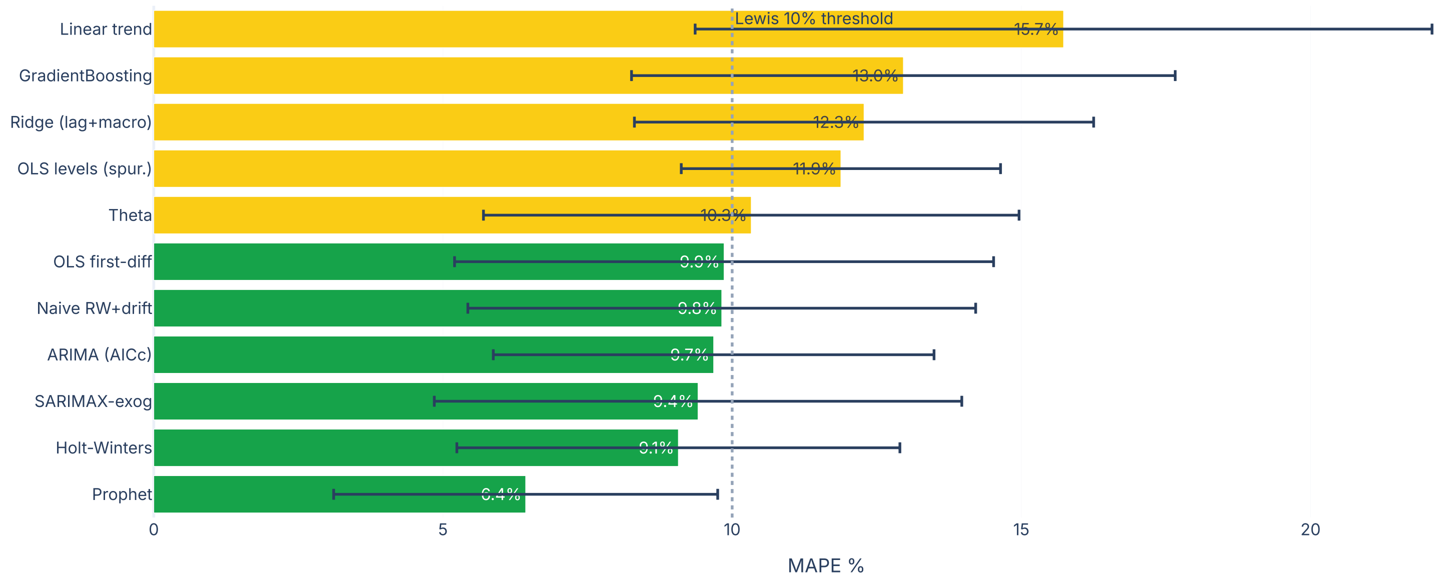
Uzbekistan — RE pipeline by oblast (bubble size ∝ installed MW)



🔬 Methodology — model scoreboard

Eleven models, 8-fold expanding-window time-series CV, pre-registered selection rule.

Forecasting model scoreboard — CV mean MAPE (error bars ±1σ across folds)



📖 Glossary

Quick reference for non-technical readers.

Units

TWh — Terawatt-hour. 1 TWh = 1 billion kWh = power for ~100,000 average households for a year.

GW / MW — Gigawatt / Megawatt — units of capacity. 1 GW = 1,000 MW.

bcm — Billion cubic metres — gas volume.

gCO_2/kWh — Grams of CO_2 emitted per kWh generated. World avg ~475; gas CCGT ~380; coal ~950; renewables ~0.

Scenarios

BAU — Business-As-Usual — current build pace; ~60% of 2030 RE target.

Government Target — Hits official April 2025 targets: 12 GW solar, 8 GW wind, 4.7 GW hydro.

Accelerated — Stretch case; >60% RE share by 2030.

Plan B (Nuclear) — Parametric SMR overlay (1.2 – 3.6 GW, 2030 – 2034).

Power-sector concepts

RE share — Hydro + solar + wind ÷ total generation. UZ ~10% (2023); target 54% (2030).

Capacity factor (CF) — Annual output ÷ nameplate. Solar 18%, wind 30%, hydro 36%, gas 55%, nuclear 85%.

T&D losses — Transmission & Distribution losses. UZ ~9-10%, regional median ~8%.

Gas self-sufficiency — Production ÷ consumption. Below 100% = net importer (UZ since 2018).

Forecasting terms

MAPE — Mean Absolute Percentage Error. Lewis 1982: <10% "highly accurate".

Cross-validation — 8 expanding-window splits × 3-year horizon.

Bootstrap CI — 80%/95% confidence band from resampling residuals 1,000×.

Hold-out — 2019-2023 chunk reserved — not used for training.

Models

Prophet — Meta's TS tool — flexible trend + changepoints. Won the bench.

ARIMA / SARIMAX — Classical statistical TS. SARIMAX adds external vars.

Holt-Winters — Exponential smoothing — robust on small samples.

Theta — Decompose-and-recombine — M3-competition classic.

OLS first-diff — Linear regression on year-on-year changes. Avoids spurious regression.

Ridge / GBM — ML on lagged + macro features. Overfit on short annual series.

Institutions

IEA — International Energy Agency (Paris).

IRENA — International Renewable Energy Agency (Abu Dhabi).

WB — World Bank — Data360 for GDP, population, CO_2.

EBRD / ADB / AIIB / IFC — Multilateral financiers active in Uzbek power.

NEEA — National Energy Efficiency Agency of Uzbekistan — ILF's key counterpart.

StatSUZ — State Statistics Committee of Uzbekistan.

Technologies

CCGT / OCGT — Combined / Open-Cycle Gas Turbine. UZ fleet mostly OCGT/CHP today.

PV — Photovoltaic — solar from semiconductor panels.

BESS — Battery Energy Storage System.

SMR — Small Modular Reactor — Plan B technology at Jizzakh.

PPP / IPP — Public-Private Partnership / Independent Power Producer.

PPA — Power Purchase Agreement — fixed long-term offtake contract.

Data: IEA · IRENA · World Bank · StatSUZ · EDB 2026. Forecasts: notebooks 03 – 06. *Toggle scenario or Plan B nuclear in the sidebar — every chart updates live.*